

Completing the finite- q complex commutative case for embeddings into quasi-Banach ℓ_p

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Status. Candidate partial result, likely valid. This packet completes the finite- q complex commutative no-embedding classification adjacent to Xu's Schatten-class question in arXiv:2207.09062. It is not a solution of the full non-commutative problem $S_q^m \rightarrow S_p^n$.

1 Source question and gap

The source paper is Chattopadhyay–Hong–Pradhan–Ray, *On Isometric Embeddability of S_q^m into S_p^n as non-commutative Quasi-Banach space*, arXiv:2207.09062 [1]. On PDF page 2, the authors record the following question, attributed to Q. Xu.

non-commutative isometric embeddability was crucial. This was also motivated from a guess of Pisier and Misra. We refer to [37] for more on this (see also [10]).

Beyond the range of $(p, q) \in [1, \infty] \times [1, \infty]$, some results are also known. By [13] for all $0 < p < q < 2$, $L_q(\mathcal{R} \overline{\otimes} B(\ell_2))$ and hence $L_q(\mathcal{R})$ embeds isometrically into $L_p(\mathcal{R})$ where $\mathcal{R}$ is a hyperfinite type II_1 factor. This generalized remarkable results due to [5] in the non-commutative setting. In [44], Sukochev and Xu studied when $L_p(\mathcal{N})$ embeds into $L_p(\mathcal{M})$ for $0 < p < 1$, where $\mathcal{M}$ and $\mathcal{N}$ are semifinite von Neumann algebras. We also refer [38] and [36] for related work. Despite these remarkable developments, the case when S_q^m embeds isometrically into S_p^n is not well studied for $0 < p \neq q \leq \infty$. The question when S_q^m embeds isometrically into S_p^n for $0 < p \neq q \leq \infty$, was asked by Q. Xu to the authors.

Question 1.1. *Let $0 < p \neq q \leq \infty$ and $m, n \geq 2$. When does S_q^m embed isometrically into S_p^n ?*

Note that [6] and [37] give partial answers to the Question 1.1 when $1 \leq p \neq q \leq \infty$. The main novelty of our work in this article is to study isometric embeddability of S_q^m into S_p^n where $0 < p \neq q \leq \infty$ as non-commutative quasi-Banach spaces which goes beyond the foregoing range of p and q . We also study the classical case, i.e. when ℓ_q^m embeds isometrically into ℓ_p^n for $m, n \geq 2$ and $0 < p \neq q \leq \infty$. Note that the case when $1 \leq p \neq q \leq \infty$ have been extensively studied in [26], [27], [28], [29], [30]. However, the authors did not consider the cases when p and q are allowed to be strictly less than 1 and to our surprise there is no more study in the literature on this. Thus our study is new even in the commutative case. Our first theorem is the following.

Theorem 1.2 (Isometric Embeddability between commutative quasi-Banach spaces $\ell_p^n(\mathbb{K})$). *Let $(q, p) \in (0, \infty) \times (0, 1)$, and $2 \leq m \leq n < \infty$ and $p \neq q$. Then there is no isometric embedding of $\ell_q^m(\mathbb{K})$ into $\ell_p^n(\mathbb{K})$ for each of the following cases:*

- (1) $\mathbb{K} = \mathbb{C}$ and $q \in (0, \infty) \setminus 2\mathbb{N}$.
- (2) $\mathbb{K} = \mathbb{R}$.

In the above theorem, the results for real and complex cases are different. The technical reason behind this is that if we consider the complex field instead of $\mathbb{R}$, that is $\ell_q^m(\mathbb{C})$, then in the proof of Theorem 1.2 it may happen that $\left. \frac{d^2}{dt^2} \right|_{t=0} \|\mathbf{a} + t\mathbf{b}\|_p^p = 0$, where $\|\mathbf{a}\|_p = \|\mathbf{b}\|_p = 1$ and in that case we can't conclude anything. Please see Remark 3.1 for more on this. These kind of facts are one of the many subtleties in our work.

Theorem 1.3 (Isometric Embeddability between non-commutative quasi-Banach spaces S_p^n). *Let $p \in (0, 1)$, and $2 \leq m \leq n$.*

- (1) *There is no isometric embedding of S_q^m into S_p^n for $q \in (0, \infty) \setminus \{1\}$ and $n < \infty$.*

The relevant source statement is Question 1.1:

Let $0 < p \neq q \leq \infty$ and $m, n \geq 2$. When does S_q^m embed isometrically into S_p^n ?

The same page states Theorem 1.2 for the commutative model $\ell_q^m(\mathbb{K}) \rightarrow \ell_p^n(\mathbb{K})$, $0 < p < 1$. In the complex case, that theorem excludes $q \in (0, \infty) \setminus 2\mathbb{N}$; hence the even positive integers $q = 2, 4, \dots$ remain outside that complex commutative statement. The new lemma below closes exactly this gap, and the following corollary combines it with source Theorem 1.2 to complete the finite- q complex commutative case.

2 Idea of the proof

The proof uses differentiability at a point selected after the embedding is fixed. Let $T : \ell_q^m(\mathbb{C}) \rightarrow \ell_p^n(\mathbb{C})$ be a hypothetical complex-linear isometry and let $f_1, \dots, f_n$ be its coordinate functionals. Because $m \geq 2$, any nonzero coordinate functional has a nonzero kernel vector x_0 . Along a line $x_0 + ty$ with $f_j(y) \neq 0$, the j -th target coordinate contributes the cusp $|tf_j(y)|^p$. This prevents differentiability at $t = 0$ when $0 < p < 1$. But the source function $x \mapsto \|x\|_q^p$ is real differentiable away from 0 whenever $q > 1$, so the isometry identity cannot hold.

3 Result

Theorem 1. *Let $0 < p < 1 < q < \infty$, and let $2 \leq m \leq n < \infty$. There is no complex-linear isometric embedding*

$$T : \ell_q^m(\mathbb{C}) \longrightarrow \ell_p^n(\mathbb{C}).$$

Proof. Assume toward a contradiction that such a complex-linear isometry T exists. Write

$$T(x) = (f_1(x), \dots, f_n(x)),$$

where $f_1, \dots, f_n$ are complex-linear functionals on $\mathbb{C}^m$. Since T is an isometry, for every $x \in \mathbb{C}^m$,

$$\sum_{r=1}^n |f_r(x)|^p = \|Tx\|_p^p = \|x\|_q^p = \left(\sum_{s=1}^m |x_s|^q \right)^{p/q}. \quad (1)$$

At least one f_j is nonzero, since T is injective. Fix such a j . Because $m \geq 2$, the kernel of f_j contains a nonzero vector; choose $x_0 \in \ker f_j \setminus \{0\}$. Also choose $y \in \mathbb{C}^m$ with $f_j(y) \neq 0$. Restrict (1) to the real line $x_0 + ty$, $t \in \mathbb{R}$:

$$\sum_{r=1}^n |f_r(x_0) + tf_r(y)|^p = \left(\sum_{s=1}^m |(x_0)_s + ty_s|^q \right)^{p/q}. \quad (2)$$

The right-hand side of (2) is differentiable at $t = 0$. Indeed, $q > 1$, so $z \mapsto |z|^q$ is real differentiable on $\mathbb{C}$, and $x_0 \neq 0$ makes the sum inside the outer power strictly positive at $t = 0$.

Now examine the left-hand side. Let

$$Z = \{r : f_r(x_0) = 0\}.$$

This set contains j , and therefore

$$c := \sum_{r \in Z} |f_r(y)|^p \geq |f_j(y)|^p > 0.$$

For $r \notin Z$, the scalar function

$$t \mapsto |f_r(x_0) + tf_r(y)|^p$$

is differentiable at 0, since its value inside the modulus is nonzero at $t = 0$. For $r \in Z$, the contribution is exactly

$$|f_r(x_0) + tf_r(y)|^p = |t|^p |f_r(y)|^p.$$

Thus the left-hand side of (2) has the form

$$h(t) + c|t|^p,$$

where h is differentiable at 0 and $c > 0$. This function is not differentiable at 0, because

$$\frac{c|t|^p}{t} = c \operatorname{sgn}(t) |t|^{p-1}$$

has no finite limit as $t \rightarrow 0$ when $0 < p < 1$. This contradicts differentiability of the right-hand side of (2). Hence no such T exists. $\square$

Theorem 2 (Complete finite- q complex commutative conclusion). *Let $0 < p < 1$, $0 < q < \infty$, $p \neq q$, and $2 \leq m \leq n < \infty$. Then there is no complex-linear isometric embedding*

$$\ell_q^m(\mathbb{C}) \longrightarrow \ell_p^n(\mathbb{C}).$$

Proof. If $q \in (0, \infty) \setminus 2\mathbb{N}$, this is exactly the complex case of source Theorem 1.2 in [1]. If $q \in 2\mathbb{N}$, then $q > 1$, and Theorem 1 applies. These two alternatives exhaust $0 < q < \infty$. $\square$

4 Scope and limitations

Theorems 1 and 2 are commutative results. They do not address the non-commutative Schatten-class embedding problem $S_q^m \rightarrow S_p^n$, where coordinate hyperplanes are replaced by spectral behavior of matrix paths and the source paper’s remaining cases require different tools. The combined conclusion also does not claim the endpoint $q = \infty$.

The direct new contribution is Theorem 1, which covers the positive even integers $q \in 2\mathbb{N}$, the finite- q values absent from the complex clause of source Theorem 1.2. Theorem 2 is the resulting complete finite- q complex commutative classification.

5 Verification and novelty check

The new proof is deterministic and uses only elementary differentiability. No computational verifier is needed.

Duplicate and novelty checks performed before promotion:

- Local cheap indexes searched: the run registry, solution index, attempt index, and proof-gap index.
- Search terms included 2207.09062, `ell_q^m`, `ell_p^n`, and isometric-embedding keywords.
- A bounded web search for exact phrases around complex $\ell_q^m \rightarrow \ell_p^n$, quasi-Banach $p < 1$, and arXiv:2207.09062 surfaced the source paper and unrelated hits, but no later explicit closing of the complex even- q commutative gap.

Human-review recommendation: check the convention that “isometric embedding” in the source commutative theorem is linear. This matches the source proof on PDF page 6, where the authors pass to a linear map $T : \ell_q^2(\mathbb{K}) \rightarrow \ell_p^n(\mathbb{K})$.

References

- [1] S. Chattopadhyay, G. Hong, C. Pradhan, and S. K. Ray, *On Isometric Embeddability of S_q^m into S_p^n as non-commutative Quasi-Banach space*, arXiv:2207.09062, 2022.